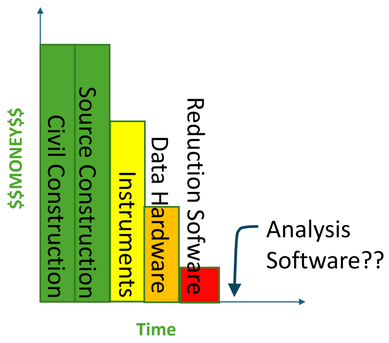
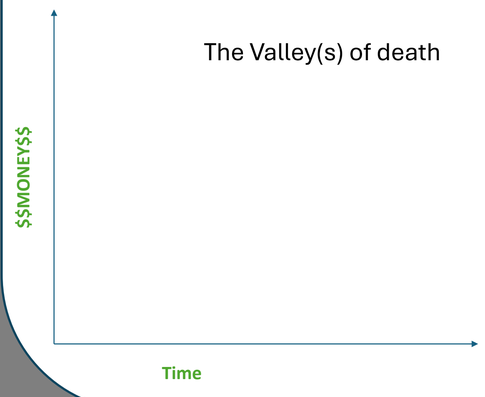
Addressing the resource problem through community: The SasView Collaboration

The Problem Statement





Analysis Software - Who's Job is it Anyway?

Scattering is an analysis tool and part of providing the tool should be the necessary software
→ the FACILITY'S JOB? Last priority with very limited funding left

Analysis is where the science is
→ the USER'S JOB? Today's analysis is beyond the skill set of most users

Or maybe …..
→ The Domain Science funding agencies?

Facility directors' discretion, NIH, NSF, DOE, etc.

Special funding (grants) do not fund long term maintenance and ongoing development. They fund "big new (transformative) ideas"

The unbounded problem

Fundamentally all the elements listed in the bar chart are relatively well-defined problems…

EXCEPT …

Analysis Software is really *unbounded*. The needs are nearly infinite and constantly evolving.

HOWEVER --- Analysis as defined here is also uniquely universal and ripe for collaborative pooling of finite resources

- A way to focus limited resources on top priorities (most useful to the science being done) in a world of infinite possibilities
- A way to harness funding for bold new ideas without losing the investment in the valleys of death
- A way to provide sustained maintenance, development and support in an uncertain funding environment (thriving through the famines)

The SasView approach: vertical and horizontal **COLLABORATIONS**

- **Facilities** provide foundational support through participation of data and instrument scientists
- **Grants and other projects** provide "bold new functionality"
- **Community**, writ large, helps provide support and functionality
- New ideas tested and developed as before by individuals or larger groups (the community)
- Once validated and deemed ready for the larger community these groups provide resources (their labor) to integrate while active developer community helps with training on where things go and on parts of interest to them (collaborative)


Camp XIII (U. of Delaware)


Camp XIV (Munich)

Open, Collaborative, Community Development

Code is open source
publicly hosted at Github
Released under BSD 3-clause license

http://github.com/SasView

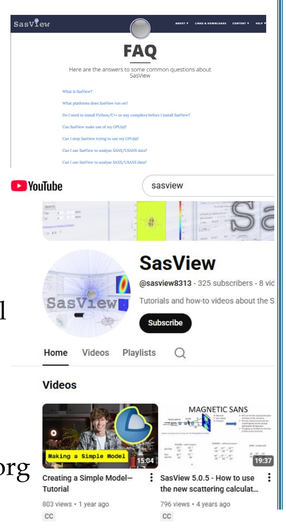
Code Hosting, **Issue Tracking,** Developer Wiki & CI on Public Github repos



Rolling **5 Year Roadmap**

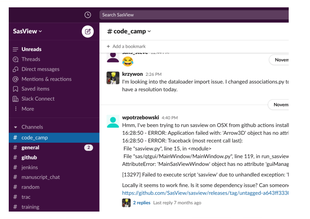
(Zenodo) DOI for **each release**

- Website http://www.sasview.org
- Documentation: in-program & online
- Written Tutorials
- Video Tutorials (YouTube)
- scattering schools/workshop
- E-learning (starting)
- Model Marketplace
- Twitter
- Bootcamps & regional workshops
- Slack
- help@sasview.org
- users@sasview.org
- developers@sasview.org



Collaboration also builds Community

- **Twice monthly Zoom calls**
- **Regular 'camps' & 'hackathons'**
- **Developer's mailing list**
- **SasView Slack**
- **Expertise sharing and helping**
- **Small leadership team to facilitate**




Hard at work together - Camp XIII


Final Sprint w/Pizza - Camp IX

Ask not what the community is going to do for you, ask what you can do for the community

No MOU … all are invited and welcome

Two Basic "Rules"

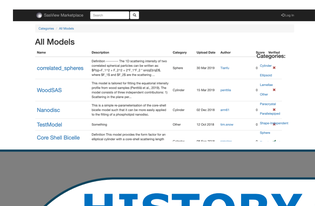
She who pays the piper chooses the tune

Those who bring the resources (time and effort, or funds to buy time and effort) choose what to work on.

And …

You cannot break existing experiences …

- New dependencies vs long term maintenance (sustainability)
  - Code quality vs long term sustainability
- Changing/degrading the current user experience for the existing user base



http://marketplace.sasview.org

Original put together by an ISIS Summer Student Lewis O'Driscoll

The SaView Package

Tools

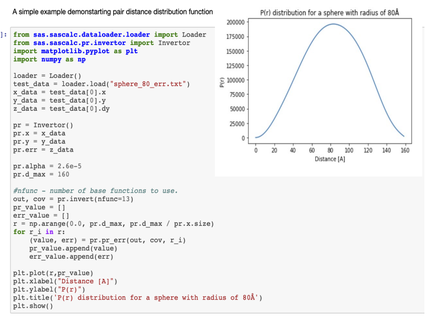
- Data Operation
- SLD calculator
- Density/Volume calculator
- Kiessig Thickness Calculator
- Q Resolution Estimator
- Orientation Viewer
- Slit Size Calculator
- Python Shell/Editor
- Image Viewer
- File Converter
- Generic Scattering Calculator
- MuMag (micromagnetics)
- Shape2SAS

Perspectives on the data



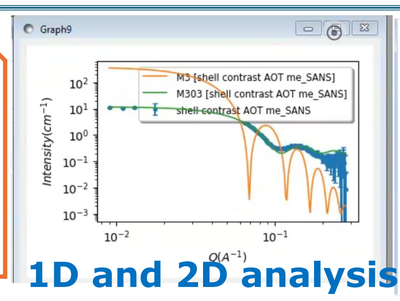
- Correlation Function
- Size Distribution
- Invariant
- P(R)
- Fitting

Jupyter Notebooks and Command Line

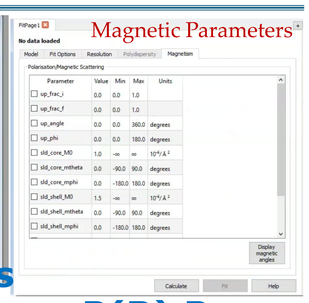


Data Management

- Most 1D formats and NXcanSAS 2D format supported.
- Some image and odd data supported through converters
- Data output in NXcanSAS or csv formats.
- Calculated curves associated with data (P(Q), S(Q), desmeared model, etc) accessible individually.
- Project saving capabilities (currently a bit fragile)

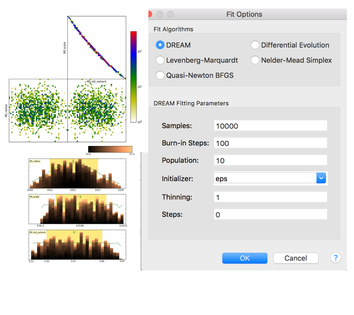


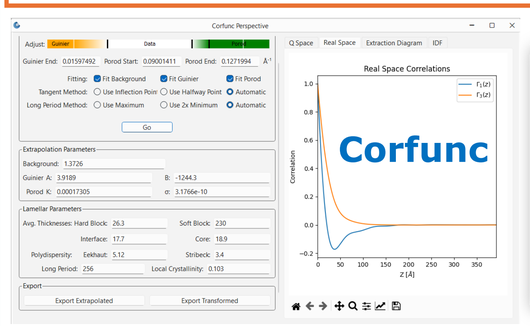
Magnetic Parameters

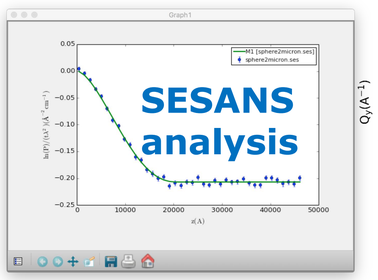


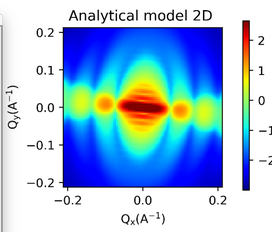
1D and 2D analysis

Choice of Optimizers

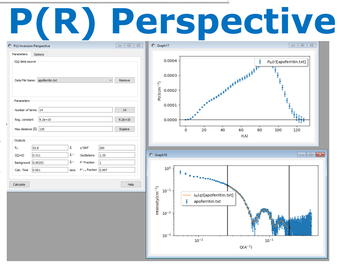




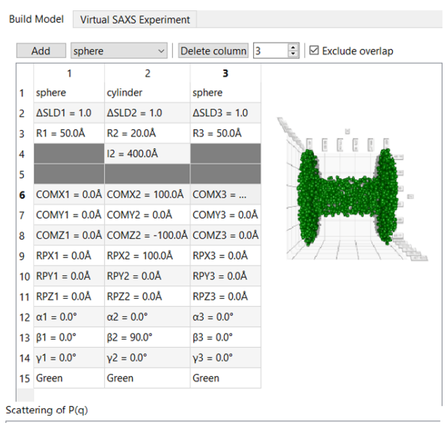




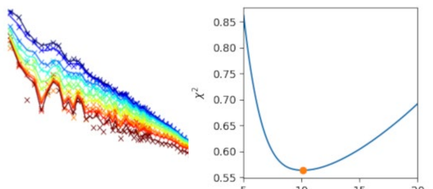
P(R) Perspective



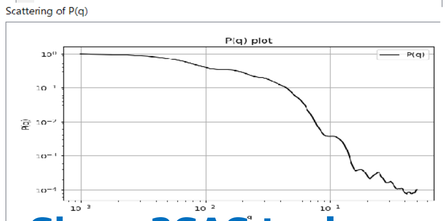
MuMag tool



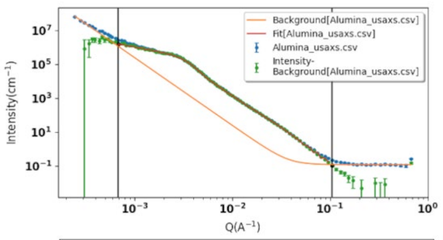
	1	2	3
1	sphere	cylinder	sphere
2	ΔSLD1 = 1.0	ΔSLD2 = 1.0	ΔSLD3 = 1.0
3	R1 = 50.0Å	R2 = 20.0Å	R3 = 50.0Å
4		l2 = 400.0Å	
5			
6	COMX1 = 0.0Å	COMX2 = 100.0Å	COMX3 = ...
7	COMY1 = 0.0Å	COMY2 = 0.0Å	COMY3 = 0.0Å
8	COMZ1 = 0.0Å	COMZ2 = -100.0Å	COMZ3 = 0.0Å
9	RPX1 = 0.0Å	RPX2 = 100.0Å	RPX3 = 0.0Å
10	RPY1 = 0.0Å	RPY2 = 0.0Å	RPY3 = 0.0Å
11	RPZ1 = 0.0Å	RPZ2 = 0.0Å	RPZ3 = 0.0Å
12	α1 = 0.0°	α2 = 0.0°	α3 = 0.0°
13	β1 = 0.0°	β2 = 90.0°	β3 = 0.0°
14	γ1 = 0.0°	γ2 = 0.0°	γ3 = 0.0°
15	Green	Green	Green

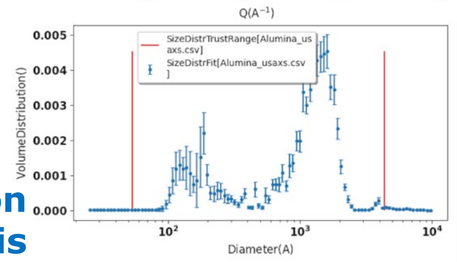


Scattering of P(q)



Shape2SAS tool





Size Distribution Analysis

HISTORY/STATUS

- 2006 - originates in NSF *DANSE* project
- 2013 - transitions into a community project
- 2016 - Funded by Sine2020 project (4 years)
- 2022 - Mostly a "volunteer army"

- 1 to 2 releases/year (6.2.0 planned for July)
- Documentation/tutorial projects ongoing
- Over the years > 100 contributors from > 30 organizations, including ~ 20 Universities across 12 countries on 3 continents (**BUT** only ~5 -15 active at any given time)
- **USAGE**: *Seems to be "everywhere" with >200 publications/year.*

SOME CHALLENGES

- The price of success: Building community resources is hard (Help IS needed!)
  - Professional looking effort leads to the misconception that the project is funded, posing a barrier to engaging community.
  - Hard to persuade non-coders of the critical importance of other contributions
- Hard to keep up with increasing security issues given resources

CURRENT PROJECTS

Major refactoring projects:

- Data loading, handling and manipulation infrastructure
- Plotting infrastructure

Major current efforts:

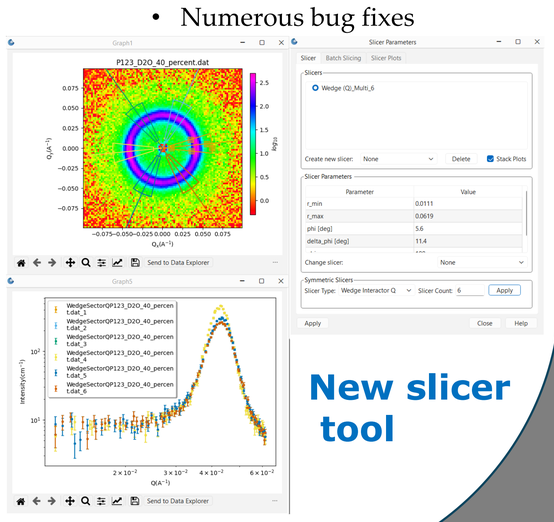
- Better support for Magnetic and polarized beam SANS and anisotropic data analysis
- Better BioSAS support and Generic Scattering Calculator
- Reimagining model fitting: Fitting 2.0


Models discussion - Camp XIV

COMING NEXT: 6.2.0

Upgraded features

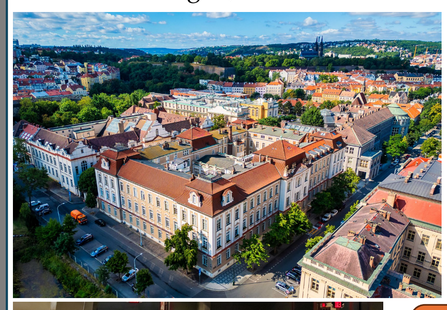
- Major slicer upgrade
- Major Invariant upgrade
- Corfunc UI upgrade
- Numerous smaller enhancements
- Numerous bug fixes



New slicer tool

CONTRIBUTOR CAMP (A Hackathon Week!)

WHEN: November 3-9, 2026 (7 days of fun!)
WHERE: Prague, Czechia at Charles University




Breaking bread together - Camp XIII

COME JOIN THE PARTY!!


Late night work at the bar- Camp XIV